

Currency Mispricing and Dealer Balance Sheets

GINO CENEDESE, PASQUALE DELLA CORTE, and TIANYU WANG

ABSTRACT

We find dealer-level evidence that recent regulation on the leverage ratio requirement causes deviations from covered interest parity. Our analysis uses a unique data set of currency derivatives with disclosed counterparty identities together with exogenous variation introduced by the U.K. leverage ratio framework. Dealers who are affected by the regulatory shock charge an additional premium of about 20 basis points per annum for synthetic dollar funding relative to unaffected dealers. This finding holds even after controlling for changes in clients' demand. Also, some clients

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increase their trading activity with unaffected dealers with whom they already had a preexisting relationship.

“We need to better understand what underpins the supply and demand that determines the basis and what causes those two curves to shift. Who are the relevant participants and what, if any, constraints might they have on their behaviour?”

— Guy Debelle (2017), Deputy Governor of the Reserve Bank of Australia

IN THE WAKE OF THE global financial crisis, policymakers and regulators embarked on a bundle of financial reforms to enhance the ability of the banking sector to absorb capital losses. As part of this package, the Basel Committee on Banking Supervision introduced a minimum requirement on the leverage ratio, defined as high-quality capital over consolidated exposures, to guard against the build-up of excessive leverage in the banking system and reduce the chances of harmful deleveraging in a downturn (BCBS (2014)). Market participants, however, have argued that this regulatory rule may have increased the costs of financial intermediation, especially for balance sheet-intensive activities, by affecting the size rather than the composition of a bank’s balance sheet relative to its capital (e.g., ESRB (2016a, 2016b)). A Reuters article on August 5, 2013, for example, argued that “[A]t the end of the day the Basel Committee has put aside some three decades of oversight based on risk-weighted assets in favour of a blunt measure of total leverage—with all kinds of unintended consequences the likely result.”

In this paper, we study the link between the leverage ratio rule for dealer banks and recent violations of covered interest parity (CIP), a no-arbitrage condition affecting foreign exchange (FX) markets (e.g., Akram, Rime, and Sarno (2008)). These deviations have made the cost of borrowing dollars through the world’s largest financial market significantly more expensive than in the U.S. money market, thus commanding a premium for international investors with dollar funding needs. While this mispricing has been associated with bank balance sheet costs that coincide with tighter leverage constraints at quarter-ends (e.g., Du, Tepper, and Verdelhan (2018)) or hedging demand (e.g., Borio et al. (2018)), the evidence is far from conclusive as existing literature relies on aggregate data that prevent a neat empirical identification of the underlying determinants.¹

Previous literature usually observes a single price in the currency market, which only allows for analysis of how a shock to capital affects the aggregate supply curve for forwards, as noted by He and Krishnamurthy (2018).

¹ This recent literature includes, among many others, Rime, Schrimpf, and Syrtstad (2017), who consider differences in liquidity premia resulting from the underlying money market rates; Arai et al. (2016), who provide evidence on the role of both monetary policy divergence and regulatory reforms as possible drivers; Andersen, Duffie, and Song (2019), who highlight the role of counterparty risk that gives rise to credit value adjustments; and Bahaj and Reis (2019), who show that central bank swap lines have created a ceiling on CIP deviations.

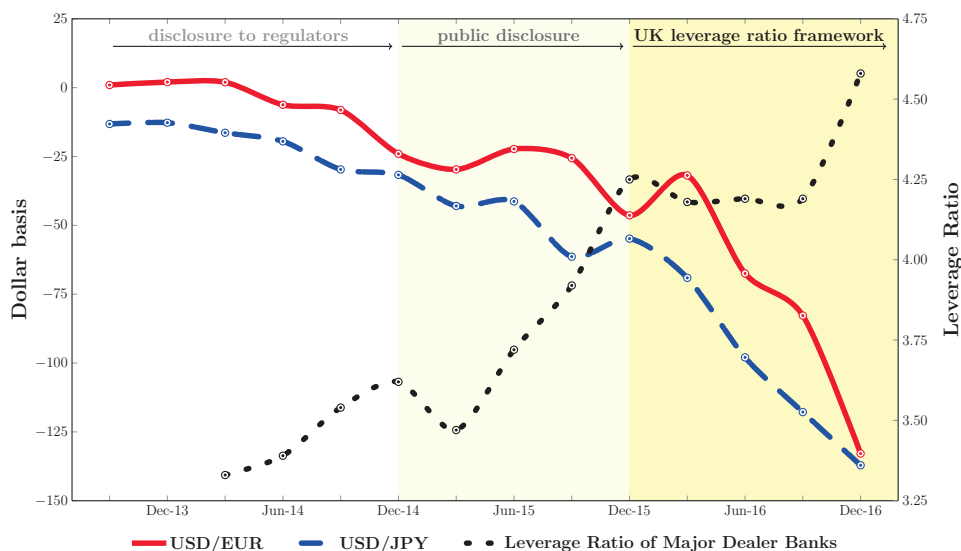


Figure 1. Dollar basis and leverage ratio. This figure displays the one-month dollar basis for the most liquid currency pairs (euro and yen against the dollar) and the average leverage ratio of major dealer banks in London (the global hub for FX trading). The dollar basis is expressed in bps per annum and computed at the end of each quarter (daily average of the last five business days) by combining LIBOR rates with spot and forward exchange rates from Bloomberg. The leverage ratio is in percentage points and is measured quarterly as high-quality capital over total asset exposure using data from the Bank of England. The public disclosure of the leverage ratio started in January 2015, whereas the U.K. leverage ratio framework for major U.K. banks was introduced in January 2016. (Color figure can be viewed at wileyonlinelibrary.com)

Simultaneous demand effects and other confounders make it difficult to tease out aggregate supply. Our paper is the first to exploit the heterogeneity across dealers and clients to control for such confounding effects and identify supply at the dealer level. As an illustrative example, Figure 1 displays the one-month dollar basis on the euro and yen against the dollar together with the average leverage ratio of major dealer banks operating in London, the global hub for FX trading. While this chart provides evidence of a strong correlation (in particular, the basis widened substantially between 2014 and 2016 while dealers faced tighter leverage constraints), establishing any causal relationship between the leverage ratio and the dollar basis is difficult as both components may be driven by common yet unobserved factors. Also, the leverage ratio requirement varies little over time and, whenever it is revised, it may change for all major banks at the same time. Finally, the cost of synthetic dollar funding may vary because of changes in both supply conditions and demand conditions and hence we should control for the latter.

To address these challenges, we exploit a quasi-natural experiment associated with the introduction of the U.K. leverage ratio framework in January 2016. This regulatory shock produces plausibly exogenous variation to

the leverage exposure of major U.K. dealer banks, thus accounting for potential time-varying omitted variables that could correlate with banks' balance sheet shocks. We employ the contract-level dollar basis with counterparty information, a critical ingredient to sweep out all time-varying counterparty- and currency-specific characteristics in the spirit of Khwaja and Mian (2008). This approach removes all confounding demand factors and is equivalent to asking whether a given counterparty in a given period that interacts with multiple dealer banks faces a larger basis from dealer banks with relatively tighter leverage conditions. Our analysis employs a novel data set on contract-level currency forwards with counterparty information and contract characteristics that corresponds to 42% of global daily trading activity (e.g., BIS (2016)). The sample covers six major currency pairs and runs between December 2014 and December 2016. We construct the contract-level dollar basis, which we then collapse into volume-weighted weekly data to identify clients with multiple trading relationships in line with Khwaja and Mian (2008).

Our identification strategy relies on a difference-in-differences method that exploits the change in the reporting obligation of the leverage ratio for major U.K. banks in January 2016.² Following a transition period of 12 months that is covered by our sample, affected banks were required to quantify their capital measure and asset exposure on the last day of each month and then average over the reference quarter. Other banks, in contrast, continued to measure their nonbinding leverage ratio on the last day of each quarter as prescribed by the European Union (EU) legislation. This shift from end-of-quarter to a monthly average reduced the ability of affected banks to window-dress their balance sheets at quarter-ends and effectively made the leverage ratio more binding. In addition, this regulatory change is presumably exogenous as major U.K. banks had no incentive to adjust their reporting obligation before its actual implementation. Our comparison shows that, on average, affected dealer banks (relative to exempted dealer banks) have charged their clients an extra premium of 16 bps per annum in response to the regulatory change.

Policymakers have argued that the persistently negative dollar basis could be attributed to the surge in FX hedging imbalances coupled with limits to arbitrage arising from tighter balance sheet constraints imposed on global banks (e.g., Arai et al. (2016), Debelle (2017), Nakaso (2017)). Divergent monetary policy stances, for example, have created an incentive for non-U.S. investors to increase their holdings of dollar-denominated assets, partly hedged for currency risk or funded by raising dollars via FX swaps. This narrative seems to suggest the existence of important heterogeneous effects across clients borrowing synthetic dollars (or selling dollars forward) and clients lending synthetic dollars (or buying dollars forward). To distinguish these effects, we employ the buy/sell indicator on forward contracts available in our data and

² This measure was part of the U.K. leverage ratio framework announced in December 2015 and made mandatory in January 2016, two years ahead of the Basel Committee proposal (Bank of England (2015b)). We thank José Manuel Marqués Sevillano (discussant) from the Banco de España for suggesting this exercise.

run difference-in-differences regressions for both types of clients. We find that affected dealer banks (relative to unaffected ones) demand a 22 bps higher premium per annum to clients borrowing synthetic dollars after the regulatory change. For clients lending synthetic dollars, in contrast, no significant arbitrage violation is recorded. To the best of our knowledge, we are among the first to show that the leverage ratio requirement affects different clients differently.

We next conduct a variety of additional exercises that confirm our main result. First, tests based on fictitious regulatory shocks lend support to the parallel trends assumption between treated and untreated dealer banks. Second, we rule out other possible changes in regulation implemented at the same time by using placebo outcome exercises based on an intramonth dollar basis or currency basis swaps, which should not be affected by the regulatory shock. Third, we examine the role of risk-weighted capital requirements by running panel regressions coupled with exogenous bank-level capital requirements. Fourth, we categorize clients as natural hedgers and other investors with the former (latter) generally borrowing (lending) synthetic dollars. We find that only natural hedgers face an additional premium. Finally, we consider the obligation for dealer banks to publicly disclose their leverage ratio after January 2015. We find that dealer banks with a low leverage ratio before the policy debate on excessive leverage (e.g., Draghi (2008)) have faced a wider negative dollar basis of 14 bps per annum relative to other dealer banks. This evidence corroborates the findings reported in our main analysis.

The impact of the leverage ratio on the dollar basis can be further amplified by the inability of dealer banks to fully offset positions on similar transactions with different clients in determining the leverage ratio. This netting restriction follows from the fact that a derivative creates an underlying asset exposure and a counterparty credit risk exposure (BCBS (2014)). We thus account for dealer-client netting activity and identify two types of clients during the pretreatment period of our exercise. The first group includes clients with zero netting against dealers (or directional clients), whereas the second one comprises clients with nonzero netting against dealers (or nondirectional clients). We find that affected dealer banks (relative to unaffected ones) have charged a significant (insignificant) premium of 54 (6) bps per annum to directional (nondirectional) clients seeking synthetic dollar funding since January 2016. Also, tighter leverage constraints may impact the trading activity of dealer banks. We extend our analysis to trading activity and document that treated dealer banks (relative to untreated banks) have reduced (increased) their volume of currency forwards by 25% (23%) against directional (nondirectional) clients selling dollars forward. For clients buying dollars forward, we continue to find no significant impact, even after accounting for dealer-client netting.

Our paper speaks to a vast literature on the validity of the CIP condition (e.g., Frenkel and Levich (1975)), which was severely compromised during the global financial crisis due to counterparty risk and U.S. dollar funding shortages (e.g., Fong, Valente, and Fung (2010), Pasquariello (2014), Ivashina, Scharfstein, and Stein (2015)). Despite an improvement in market conditions, recent years have again been characterized by large and systematic arbitrage

deviations. For example, Du, Im, and Schreger (2018) measure CIP deviations between U.S. Treasury bonds and foreign bonds in local currencies and document a secular decline at medium to long maturities. Du, Tepper, and Verdelhan (2018) document that one-week and one-month CIP deviations are more pronounced at quarter-ends after January 2015, when banks started to publicly disclose their leverage ratio. Eguren-Martin, Ossandon Busch, and Reinhardt (2018) find that banks reduce their supply of cross-border credit in a particular currency when the cost of obtaining funding through that currency increases. We go beyond this literature by identifying supply at the *dealer level*, exploiting more variation in the data arising from the introduction of the U.K. leverage ratio framework, and controlling for changes in client demand.

Our findings suggest that financial intermediaries play an important role in the pricing of financial assets (Brunnermeier and Pedersen (2009), He and Krishnamurthy (2013), Adrian, Etula, and Muir (2014), He, Kelly, and Manela (2017), Du, Hébert, and Huber (2019)). In the context of the FX literature, Gabaix and Maggiori (2015) present a theory of exchange rate determination that builds on the limited risk-bearing capacity of intermediaries and links several stylized facts to intermediaries' balance sheets. Our results are also related to the literature on limits to arbitrage. Gromb and Vayanos (2018) propose a model in which arbitrageurs have limited capital, which constrains their activity and gives rise to multiple pricing anomalies. More recently, Hazelkorn, Moskowitz, and Vasudevan (2020) find that the bases between futures and spot prices are driven by intermediaries' cost of capital and are consistent with uninformed leverage demand. Finally, we contribute to the literature on the impact of financial regulation on asset markets (e.g., Adrian, Boyarchenko, and Shachar (2017), Cenedese, Ranaldo, and Vasios (2020)). Kotidis and van Horen (2018) show that the leverage ratio negatively affects dealer-client repo intermediation.

The remainder of this paper is organized as follows. Section I reviews the CIP condition and recent regulatory changes. Section II provides a detailed description of the data and explains the construction of the contract-level dollar basis. Section III exploits the introduction of the U.K. leverage ratio framework for the dollar basis, while Section IV provides further analysis and refinements of the main results. Section V concludes. A separate [Internet Appendix](#) provides additional robustness tests and supporting analysis.³

I. Background

In this section, we begin by reviewing CIP. We then summarize the key aspects of postcrisis financial regulation, with a particular emphasis on the U.K. regulatory framework.

³ The [Internet Appendix](#) may be found in the online version of this article.

A. A Review of Covered Interest Rate Parity

The gap between the funding cost in the U.S. cash market and the funding cost implied through the FX market is referred to as the *dollar basis*. When CIP holds, the dollar basis is zero. In contrast, when CIP breaks down, the dollar basis deviates from zero and arbitrage opportunities arise. As an example, suppose that the dollar basis is negative. Then an arbitrageur could earn a profit by simultaneously borrowing in the dollar cash market, selling dollars for foreign currency in the spot market, lending the resulting amount in the foreign cash market, and entering a forward contract to convert the final proceeds in dollars at maturity, thus offsetting the initial short domestic position. This strategy is often implemented via an FX swap contract, which combines a spot with a forward transaction.

The CIP no-arbitrage relationship is routinely summarized as follows :

$$1 + r_{\ell,t} = \frac{F_{i\ell,t}}{S_{i,t}}(1 + r_{i\ell,t}),$$

where $S_{i,t}$ denotes the spot exchange rate expressed in units of dollars per foreign currency i , $F_{i\ell,t}$ is the corresponding forward exchange rate with delivery date $t + \ell$, and $r_{\ell,t}$ ($r_{i\ell,t}$) is the (foreign) interest rate at time t with maturity ℓ . While the left-hand side of this equation indicates the gross return from lending in the dollar cash market, the right-hand side is the swap-implied dollar rate obtained from lending in the foreign cash market while *covering* foreign currency risk. When the underlying cash market instruments are equivalent in all respects except the currency of denomination, both strategies are equivalent and thus must deliver an identical payoff. This is the usual textbook-style exposition, which abstracts from bid and ask prices, assumes an absence of default, liquidity, counterparty, and settlement risk, and relies on periodic interest rates (e.g., Bekaert and Hodrick (2012)).

Since mid-2014, the FX market has experienced a large and persistent negative dollar basis as documented by Du, Tepper, and Verdelhan (2018), among many others. In this paper, we empirically show that such a negative basis can be attributed to a steeper supply curve of forwards resulting from tighter leverage ratio constraints. As an illustrative example in the spirit of Prachowny (1970), suppose that a Japanese investor borrows an amount x_i in the local money market, exchanges yen for dollars in the spot market, invests in the U.S. money market, and hedges her currency exposure by selling dollars for yen in the forward market. Assume that the supply of yen in the forward market is such that the forward price rises with the quantity of funds transacted x_i , that is, $dF_i/dx_i > 0$.

The dealer takes the other side of the transaction. Suppressing subscripts ℓ and t for simplicity, his profit, π , is defined as

$$\pi = x(1 + r_i) \frac{F_i}{S_i} - x(1 + r),$$

with profit maximization implying that

$$\frac{\partial \pi}{\partial x_i} = (1 + r_i) \frac{F_i}{S_i} + (1 + r_i) x_i \frac{dF_i}{dx_i} - (1 + r) = 0. \quad (1)$$

By rearranging equation (1), we obtain

$$1 + r = (1 + r_i) \left(\frac{F_i}{S_i} + \frac{F_i}{E_s} \right), \quad (2)$$

where $E_s \equiv (dx_i/dF_i)(F_i/x_i)$ denotes the elasticity of the supply of forwards. With perfectly elastic supply (i.e., $E_s = \infty$), equation (2) reduces to the textbook no-arbitrage condition. When the supply of forwards becomes inelastic (i.e., $E_s \rightarrow 0$), the dollar basis turns negative. We next outline the key components of the financial regulation introduced in the aftermath of the global financial crisis.

B. Postcrisis Financial Regulation

Since the global financial crisis, policymakers and regulators have introduced a significant program of financial reforms to strengthen the resilience of the banking sector and reduce negative spillovers from the financial sector to the real economy. Chief among these reforms is the new regulatory framework on bank capitalization, stress testing, and market liquidity risk announced by the Basel Committee on Banking Supervision in July 2010. This new framework is generally referred to as the “Basel III” accord (e.g., BCBS (2009, 2010)).

As part of this comprehensive bundle of rules, the Basel Committee proposed a nonrisk-weighted capital requirement—the leverage ratio—according to which a bank has to hold a minimum level of high-quality loss-absorbing (Tier 1) capital in proportion to its total consolidated assets. The latter include on-balance-sheet exposures, derivatives exposures, securities financing transactions exposures, and other off-balance-sheet items (e.g., BCBS (2014)).⁴ By weighting all exposures equally, the leverage ratio can potentially increase the intermediation costs of those assets that are characterized by a low margin and high volume. This is especially true for FX derivatives, which can substantially expand a dealer’s balance sheet and thus attract a capital charge under the leverage ratio framework. Moreover, for the calculation of the leverage ratio, banks have limited ability to net out derivatives exposures that offset each other across different counterparties, as derivatives generate underlying asset exposure and counterparty credit risk exposure (e.g., BCBS (2014)).

In May 2014, the Basel Committee finalized the definition of the leverage ratio (BCBS (2014)) and required that banks publicly disclose their leverage ratio on a quarter-end basis starting in January 2015; compliance with minimum leverage ratio requirement was postponed to January 2018. In

⁴ The leverage ratio differs from the capital ratio, which is defined as capital over risk-weighted assets. The weights depend on both credit risk and market risk, with the latter measured using a value-at-risk (VaR) based on 10-day holding-period returns.

the United Kingdom, however, a binding leverage ratio requirement was announced by the Prudential Regulation Authority in early December 2015 and became mandatory for major U.K. banks (those with deposits over £50 billion) in January 2016. The U.K. leverage ratio framework consists of a *minimum leverage ratio requirement* of 3% (three quarters of which must be met with common equity Tier 1 capital instruments), a *countercyclical buffer*, and an *additional buffer for global systemically important institutions* (Bank of England (2015b, 2015c)). With respect to the reporting requirements, the regulator introduced a transition period of 12 months during which the numerator and the denominator of the leverage ratio were calculated using end-of-month figures during the reference quarter. Other regulated banks, that is, small domestic banks as well as the subsidiaries of foreign banks, remained subject to the requirements set by the Basel Committee and continued to report their leverage ratio as a point-in-time estimate at quarter-end periods.⁵ The key dates are presented in Figure IA.1 in the Internet Appendix.

In this paper, we exploit the two regulatory events above, namely, the introduction of the U.K. leverage ratio framework in 2016 (main analysis) and the public disclosure of the leverage ratio in 2015 (robustness analysis) to conduct difference-in-differences regressions. This exercise helps improve our identification, and thus offers a supply-side explanation for the existence of a nonzero dollar basis.

II. Data Description

In this section, we first describe the trade repository data on forward contracts. We then describe the construction of the contract-level dollar basis that we use in our empirical analysis.

A. Trade Repository Data

Over-the-counter (OTC) derivatives markets are regarded as the most opaque financial markets. Understanding the complexity and functioning of these “dark markets” is notoriously difficult as buyers and sellers negotiate the terms of the trade privately (e.g., Duffie (2012)). Not surprisingly, financial regulators have long struggled to gather key information such as price, volume, maturity, outstanding transactions, and counterparty identities. The recent global financial crisis, however, marked an important turning point. In particular, in response to the crisis, in September 2009 G20 leaders put forward a broad reform agenda to improve the level of transparency in these markets. As part of this initiative, it was agreed that all OTC derivatives contracts would be reported to trade repositories to grant policymakers and regulators access to high-quality, high-frequency data.

⁵ In the EU, Basel III was introduced through a legislative package covering prudential rules for banks, building societies, and investment firms, namely, the Capital Requirements Directive (2013/36/EU) implemented through national law and the Capital Requirements Regulation (575/2013) directly applicable to firms across the Union.

In the EU, this commitment was introduced with the European Market Infrastructure Regulation (EMIR), which requires that EU residents report details of derivative transactions to a trade repository authorized by the European Securities and Markets Authority (ESMA) by the next business day. This reporting obligation covers all asset classes and applies to clearinghouses, financial counterparties, and nonfinancial counterparties that are legal entities under the EU jurisdiction. While this reporting obligation was introduced in February 2014, a large number of observations were initially missing or incorrectly reported. In response, ESMA introduced a formal data validation process in November 2014 that has substantially improved the quality of the data (see Abad et al. (2016)).

We have been given access to EMIR transactions, in which at least one of the counterparties is a U.K. legal entity, by the Bank of England. Our data set consists of currency forwards (i.e., outright forwards and forward legs of FX swaps) and currency basis swaps (i.e., floating-for-floating currency swaps) traded between December 2014 and December 2016 on six major currency pairs, namely, Australian dollar (AUD), Canadian dollar (CAD), Swiss franc (CHF), euro (EUR), British pound (GBP), and Japanese yen (JPY) relative to the U.S. dollar (USD). We rely on submissions to DTCC Derivatives Repository, the largest trade depository in terms of market share, given a lack of data harmonization across trade repositories. For each transaction, we observe counterparty information (i.e., legal identifier and corporate sector) and contract characteristics (e.g., price, notional amount, maturity date, execution date, and execution time) for more than 100 fields. We uncover an average daily volume of \$867 (\$24) billion for currency forwards (currency basis swaps), which corresponds to more than 42% of the daily trading activity for both sets of instruments (e.g., BIS (2016), consistent with the fact that London is the world's largest FX trading center. See Figures IA.2 and IA.3 in the [Internet Appendix](#) for additional details.

We collect raw data from the “trade activity report” and discard duplicates of the same transaction using the unique trade identifier, as the EMIR imposes a double-sided reporting regime. In some cases, we remove multiple copies of the same trade due to modifications, corrections, and valuation updates. Finally, we drop transactions missing key information and we remove observations with extreme notional values by winsorizing the data at the 99.9% level.

The FX market consists of two tiers: an interbank market where dealers (typically large international banks) trade among themselves and a retail segment where financial and nonfinancial clients trade with either dealers or other clients. In line with this characterization, we categorize individual counterparties into dealers and clients and then group their transactions accordingly. We classify as dealers the largest banks by market share according to the 2015 and 2016 Euromoney FX survey. Using this criterion, we end up with a list of 17 dealers, which comprises (in alphabetic order) Bank of America Merrill Lynch, Barclays, BNP Paribas, Citi, Crédit Agricole, Credit Suisse, Deutsche Bank, Goldman Sachs, HSBC, JP Morgan, Morgan Stanley, Nomura, Royal Bank of Scotland, Société Générale, Standard Chartered,

State Street, and UBS. This set of dealers is obtained by consolidating up to 106 (51) different legal entities for currency forwards (currency basis swaps). Dealers' counterparties are manually separated into several sectors, namely, real money investors (asset managers, pension funds, insurance firms, state institutions, and unclassified funds), hedge funds, nonfinancial corporates, nondealer banks (commercial banks, prime brokerage firms, and nonbank firms offering trading services), and central banks (including monetary authorities). In our sample, we classify more than 30,000 (400) clients for currency forwards (currency basis swaps), which correspond to more than 98% of total trading volume. The residual 2% is associated with unclassified clients, in particular, individual counterparties for which the legal identifier was missing/incorrect or associated with an entity difficult to categorize. A breakdown by currency, maturity, and sector is provided in [Internet Appendix Section I](#).

B. Contract-Level Dollar Basis

We measure the dollar basis by synchronizing contract-level currency forwards with midquotes on spot exchange rates and overnight index swap (OIS) rates from Refinitiv Tick History at the second level.⁶ Depending on the availability of OIS rates, we use all forwards between a one-week and three-month maturity. For nonstandard maturity contracts (e.g., a 45-day forward), we rely on linearly interpolated OIS rates using the closest available tenors (e.g., one-month and two-month OIS rates). For AUD, CHF, and CAD, we restrict the analysis to maturities between one month and three months as OIS rates are not available for shorter maturities.

We construct the dollar basis for each contract τ , for nearly 3.5 million observations, according to

$$B_{ijk\ell,\tau} = \left[(1 + r_{\ell,\tau}) - (1 + r_{i\ell,\tau}) \frac{F_{ijk\ell,\tau}}{S_{i,\tau}} \right], \quad (3)$$

where $B_{ijk\ell,\tau}$ denotes the dollar basis for currency i , dealer j , counterparty κ , and maturity ℓ , $r_{\ell,t}$ ($r_{i\ell,t}$) is the dollar (foreign) interest rate with maturity ℓ , $S_{i,t}$ is the spot exchange rate defined in units of dollars per currency i , and $F_{ijk\ell,\tau}$ is the corresponding forward exchange rate. Only the latter price differs across parties consistent with the view that a nonzero basis is primarily driven by shifts in the supply and/or demand of forwards (Du, Tepper, and Verdelhan (2018), Borio et al. (2018)). Finally, we appropriately scale the interest rates (quoted in percentage per annum) and express $B_{ijk\ell,\tau}$ in bps per annum.

In our analysis, the forward price can be a bid or an ask price whereas interest rates and spot exchange rates are midquotes. We abstract from

⁶ The dollar basis is generally computed with unsecured interbank rates (e.g., LIBOR rates based on quotes submitted at 11:00 a.m. London time) coupled with spot and forward exchange rates (e.g., prices recorded by Bloomberg at 6:00 p.m. London time). [Internet Appendix Table IA.I](#) shows that replacing LIBOR with OIS rates sampled at 11:00 a.m. London time generates a qualitatively similar dollar basis.

Table I
Descriptive Statistics: Dollar Basis

This table presents descriptive statistics for the dollar basis measured at the currency, dealer, and client level in bps per annum. Panel A employs maturities between one week and three months. Panel B (C) uses maturities between one week and one month (longer than one month). Q_{25} , Q_{50} , and Q_{75} denote the first, second, and third quartile, respectively. All statistics are first calculated daily using the contract-level dollar basis and then averaged across the entire sample. The dollar basis is constructed by synchronizing contract-level currency forwards between dealers and clients from DTCC Derivatives Repository with spot exchange rates and interest rates from Refinitiv Tick History between December 2014 and December 2016.

Panel A: All Maturities							
	Mean	Standard Deviations			Q_{25}	Q_{50}	Q_{75}
		Overall	Between Dealer	Within Dealer			
AUD	−7.93	86.45	32.42	78.84	−49.14	−3.23	38.25
CAD	−18.42	77.82	32.83	69.21	−49.06	−13.87	19.60
CHF	−68.47	69.53	27.40	62.51	−102.93	−68.44	−29.66
EUR	−29.38	87.91	21.54	84.69	−54.84	−31.03	0.96
GBP	−17.65	94.13	24.62	90.13	−51.52	−18.30	20.13
JPY	−46.03	90.30	24.78	86.22	−81.47	−47.02	−5.91
Panel B: Short Maturities							
EUR	−22.75	102.12	29.46	96.80	−57.70	−22.11	22.78
GBP	−16.82	109.26	34.05	102.60	−62.98	−16.62	37.33
JPY	−39.74	103.23	33.96	96.57	−83.51	−36.56	14.05
Panel C: Longer Maturities							
EUR	−35.33	70.27	20.61	66.58	−53.29	−35.32	−15.29
GBP	−17.68	75.40	22.22	71.33	−43.49	−19.07	8.95
JPY	−54.49	68.45	23.56	63.46	−80.14	−54.57	−27.24

bid-ask spreads as Du, Tepper, and Verdelhan (2018) show that transaction costs cannot fully explain CIP deviations in the recent period. Moreover, we are interested in what causes the basis to move rather than whether the basis represents a “free lunch.” In particular, because our objective is to link the dollar basis to the leverage ratio constraints of dealer banks, our analysis focuses on transactions between dealers and clients. This focus allows us to control for unobserved characteristics at the client level in the spirit of Khwaja and Mian (2008) and thus isolate demand and supply factors.

Table I reports descriptive statistics for the contract-level dollar basis in bps per annum. We first construct means, standard deviations, and quartiles for each business day using forward transactions between dealers and clients within the same day. We then average these measures across the entire sample. We employ maturities between one week and three months in Panel A, between one week and one month in Panel B, and longer than one month in Panel C. Because of data availability, we only display the maturity

decomposition for the major currency pairs. The dollar basis is generally negative for all currency pairs and increases, in absolute terms, with the maturity of the forward contracts, in line with Du, Tepper, and Verdelhan (2018). In Panel A, for example, the average basis for the USD/JPY is about -46 bps per annum. Clients at the 75th percentile pay a basis close to -6 bps per annum and clients at the 25th percentiles face a basis below -81 bps per annum. This basis widens with the maturity as we document an average basis of -40 and -54 bps per annum, respectively, in Panels B and C.

We observe a substantial amount of dispersion, as measured by the standard deviation, ranging between 70 and 94 bps for USD/CHF and USD/GBP when all maturities are pooled together. Since our granular data set captures heterogeneity across both dealers and clients, it is natural to ask whether this source of variation reflects dispersion that occurs between or within dealers. To shed light on this question, we calculate between-dealer and within-dealer dispersion. Focusing first on within-dealer dispersion, it is fairly high and ranges between 63 bps for the USD/CHF and 90 bps for USD/GBP in Panel A, and it is about 13% higher for short-maturity dollar basis. These findings are consistent with recent literature that documents large price dispersion in OTC markets, which are characterized by imperfect competition, opacity, heterogeneous clientele, price discrimination, bargaining, and search costs for investors (e.g., Duffie, Gârleanu, and Pedersen (2005, 2007), O'Hara, Wang, and Zhou (2018), Hendershott et al. (2020)). For FX markets in particular, Bahaj and Reis (2019) provide theoretical arguments suggesting that a different degree of bargaining power between dealers and traders over the terms of a forward contract can produce bank-specific dollar basis. Similarly, Hau et al. (2019) use contract-level forwards on the USD/EUR between dealers and nonfinancial corporate clients and document large price dispersion, which they attribute to search frictions and client sophistication.

We also document substantial between-dealer dispersion after controlling for residual within-dealer variation, meaning that dealers can charge a different dollar basis to their clients. Specifically, between-dealer dispersion ranges between 22 bps for the EUR and 33 bps for the CAD in Panel A, and its cross-currency average is around 32 bps in Panel B and 22 bps in Panel C. A possible explanation for these results could be that dealer banks face different capital constraints and thus different shadow costs, in the spirit of the intermediary asset pricing literature (e.g., He and Krishnamurthy (2013)). In the insurance market, for example, one can observe different transaction prices for the same asset sold by insurers to households. Koijen and Yogo (2015) find that life insurers sold policies at prices below actuarial fair value and that this gap was larger for insurers facing tighter capital constraints. In the FX literature, Bahaj and Reis (2019) show that regulatory requirements affecting dealer banks constitute an additional source of variation for the distribution of the dollar basis across dealer banks, and Abbassi and Bräuning (2021), who study contract-level forwards on the USD/EUR between German banks and their counterparties, find large cross-sectional dispersion for forward premia, which they attribute to the banks' shadow cost of capital rather than the type

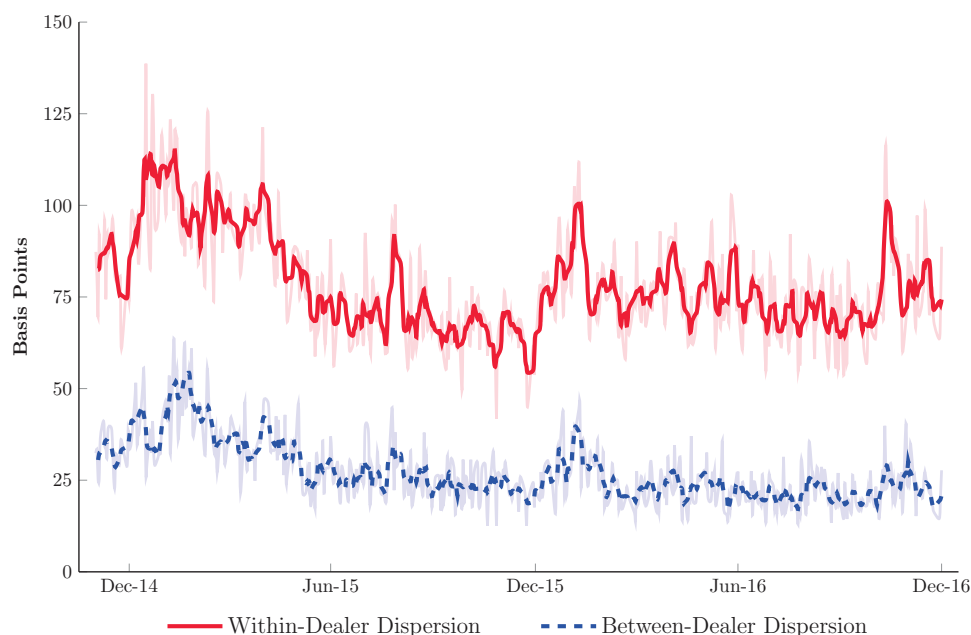


Figure 2. Decomposing the dispersion of the dollar basis. This figure displays the daily within-dealer and between-dealer dispersion (or standard deviation) of the dollar basis. Dispersion is calculated first using the dollar basis measured at the currency, dealer, and client level in bps per annum and then averaged across all currency pairs. The shaded lines denote the daily standard deviation whereas the solid lines show the corresponding five-day centered moving average. The dollar basis is constructed by synchronizing currency forwards between dealers and clients from DTCC Derivatives Repository with spot exchange rates and interest rates from Refinitiv Tick History. The sample runs from December 2014 to December 2016. (Color figure can be viewed at wileyonlinelibrary.com)

of bank or contract (e.g., small or domestically focused banks, client trades, etc.).⁷ Figure 2 depicts between-dealer and within-dealer dispersion computed on each business day. We use currency forwards with a maturity between one week and three months and then average across all currency pairs. We plot these quantities (shaded lines) as well as the corresponding five-day centered moving averages (solid lines), which mitigate, for example, day-of-week effects. As can be seen, both within-dealer and between-dealer dispersion display a fair amount of variation over time, with a peak at the beginning of 2015 and 2016. Overall, the existence of cross-dealer dispersion means that sufficient variation exists in the forward market to empirically measure the impact of regulatory restrictions on dealer banks. We thus complement the existing literature, which typically relies on one forward price and as a result can only

⁷ Table IA.II in the [Internet Appendix](#) presents within-dealer and between-dealer dispersion after controlling for dealer and client sector fixed effects. The results remain qualitatively similar.

provide insights into how shocks to capital impact the aggregate supply curve for forwards, as He and Krishnamurthy (2018) point out.

C. Effective Spread on Forwards

We also compute the effective spread for currency forwards using our transaction prices synchronized with midquotes from Refinitiv Tick History, similar to Hau et al. (2019). Specifically, we construct the effective spread in relative terms, to ease comparison across currency pairs, for each transaction τ as

$$E_{ijk\ell,\tau} = d_\tau \times \left(\frac{F_{ijk\ell,\tau} - F_{i,\tau}}{F_{i,\tau}} \right), \quad (4)$$

where $F_{ijk\ell,t}$ is the contract-level forward, $F_{i,\tau}$ is the forward midquote, and d_τ is the trade direction indicator associated with $F_{ijk\ell,t}$, that is, $d_\tau = 1$ when a client buys foreign currency (or equivalently sells U.S. dollars) in the forward market and $d_\tau = -1$ when a client sells the foreign currency (or equivalently buys U.S. dollars) in the forward market. We express the effective spread in bps. For this exercise, we linearly interpolate forward midquotes across the closest available tenors for nonstandard tenors and rely on the buy/sell indicator available in our data set for currency forwards between October 2015 and December 2016, for an overall sample of about two million observations.

We present descriptive statistics on the relative effective spread in Table IA.III in the Internet Appendix. Similar to Table I, we first construct means, standard deviations, and quartiles for each business day using all dealer-to-clients forward transactions within a given day, and then average them across the entire sample. The average spread is generally below one bp, increases with contract maturity, and displays large dispersion.⁸ We also report the between-dealer and within-dealer dispersion, which are proportionally similar to those recorded in Table I. For example, the average within-dealer dispersion is approximately 10 bps while the average within-dealer dispersion is not far from 3 bps, thus implying a substantial amount of cross-dealer dispersion, after accounting for the residual dealer dispersion. Overall, this exercise suggests that a large amount of dispersion observed in our data set is a genuine feature of the data rather than an artifact of our matching procedure.⁹

⁸ If we restrict our sample to currency forwards on the USD/EUR between dealers and corporate clients and use the effective spread in pips (i.e., the numerator of equation (4)), the mean is about 5.7 pips (with a standard deviation of 28.6), which is largely comparable to the mean of 6.9 pips (with a standard deviation of 19.4) reported in Hau et al. (2019).

⁹ Table IA.IV in the Internet Appendix presents summary statistics for the forward premia computed by synchronizing contract-level forwards with spot rate midquotes from Refinitiv Tick History. Both within-dealer and between-dealer dispersion are proportionally comparable to those reported in Table I.

III. Dollar Basis and Leverage Ratio

In this section, we study the impact of the regulatory leverage ratio on the dollar basis, thus offering a supply-side interpretation for the recent failure of the CIP condition. We exploit two plausibly exogenous sources of variation affecting the leverage exposure of major dealer banks within a quasi-natural experimental exercise. The first source of exogenous variation, which we use in our main analysis, is based on the introduction of the U.K. leverage ratio framework in January 2016. The second, which we use in a robustness exercise, is the public disclosure of the leverage ratio in January 2015. Following the path-breaking work of Khwaja and Mian (2008), we enhance our identification strategy by absorbing changes in demand conditions at the client $\times$ currency level by including a client-currency-time fixed effect. For its implementation, we collapse our contract-level dollar basis into a weekly volume-weighted dollar basis as this allows us to identify clients with multiple trading relationships. This approach removes all confounding demand factors and is equivalent to asking whether a given counterparty in a given period interacting with multiple dealer banks faces a larger basis from the dealer bank with a relatively tighter leverage constraint.

A. The U.K. Leverage Ratio Framework

In our first difference-in-differences exercise, a plausibly exogenous shock arises from the introduction of the U.K. leverage ratio framework in January 2016, which only affected major U.K. banks. The subsidiaries of foreign banks (including all other domestic banks) continued to fall under the prescription of the EU legislation. This regulatory change constitutes an ideal quasi-natural experiment to identify the impact of the leverage ratio rule on the dollar basis as it allows us to identify treatment and control groups of dealer banks and thereby conduct difference-in-differences analysis. In our sample, four dealer banks (Barclays, HSBC, Royal Bank of Scotland, and Standard Chartered) form our treatment group, and the subsidiaries of international banks (Citibank, Credit Suisse, Goldman Sachs, JP Morgan, Morgan Stanley, and Nomura) act as a control group. While the selection within each block is not random, all of these banks are categorized as global systemically important banks (G-SIBs) by the Financial Stability Board and the Basel Committee and thus should be regarded as largely comparable. This is important as causal inference would be compromised if our treated banks were systematically different, in terms of, say, business models or funding strategies, from control banks.¹⁰

In addition to a minimum requirement of 3% plus additional buffers, the U.K. framework introduced an important change to the reporting obligations. During a transition period of 12 months from January to December 2016, major U.K. banks were required to quantify the key ingredients for the calculation of

¹⁰ Other European dealer banks are not included in our control group as they have branches rather than subsidiaries in the United Kingdom. As a result, they are not U.K. legal entities and most of their dealer-to-client transactions are not observed by the U.K. regulator.

the leverage ratio (i.e., capital and exposure measures) on the last day of each month during the reference calendar quarter and then average them. In contrast, other banks continued to report the leverage ratio recorded on the last day of each quarter. This end-of-month average rule aimed to reduce regulated banks' ability to window-dress their balance sheet at quarter-ends, and thus make the leverage ratio rule more effective. As banks had little incentive to adjust their reporting obligations ahead of the implementation date, the change in the reporting obligations can be seen as plausibly exogenous variation that affected one group of banks while leaving other similar banks unaffected.

In our analysis, the preintervention period runs from November 2 to December 18, 2015, whereas the postintervention period runs from January 11 and February 26, 2016. Both periods are intended to capture two months of observations before and after the regulatory shock while removing the weeks bracketing the year-end to avoid, for example, the G-SIB surcharge, end-of-year volatility, and tax-driven window-dressing effects (e.g., Kotidis and van Horen (2018)). Our results remain both quantitatively and qualitatively similar if we retain these two weeks of data. For this analysis, we also discard forward contracts with intramonth contractual and maturity dates as these contracts should not affect the balance sheet exposure of dealer banks at the end of the month. An example would be a short-maturity forward contract agreed on November 13, 2015 and expiring on November 20, 2015. Moreover, to minimize any cross-period contamination, we further eliminate forward contracts traded during the preintervention period and affecting the balance sheet exposure of a dealer bank at the end of the month during the postintervention period. For instance, a two-month forward contract traded on December 15, 2015 (preintervention period) would affect the balance sheet of a dealer at the end of January 2016 (postintervention period).

B. Methods and Empirical Evidence

We use difference-in-differences regressions based on the specification

$$B_{ijk,t} = \gamma D_{affected} \times D_{post} + \delta' X_{j,t-1} + fe + \varepsilon_{ijk,t}, \quad (5)$$

where $B_{ijk,t}$ is the volume-weighted dollar basis measured in week t for currency i , dealer j , and counterparty κ , $D_{affected}$ is a dummy variable that takes the value of one for affected dealers and zero for unaffected dealers and thus captures the time-invariant difference in outcomes between the two groups, D_{post} is a dummy variable that selects observations after the regulatory intervention and absorbs the impact of time-varying factors that are group-invariant, $D_{post} \times D_{affected}$ is an interaction term between the time and treatment group dummies, and $X_{j,t-1}$ is a set of lagged control variables that capture time-varying observed dealer characteristics at the weekly frequency, for example, counterparty and liquidity risks. We use the first-difference of the five-year CDS spread and the log volume of currency forwards. The coefficient γ measures the treatment effect, that is, the difference in the dollar

Table II
Dollar Basis and Leverage Ratio

This table presents difference-in-differences estimates associated with the introduction of the U.K. leverage ratio framework on January 1, 2016. The dependent variable is the volume-weighted dollar basis from currency forwards measured weekly at the currency, dealer, and client level in bps per annum. *Affected Dealers* is a dummy equal to one (zero) for major U.K. banks (subsidiaries of international banks). *Post Regulation* is a dummy that equals one (zero) for a posttreatment (pre-treatment) period of two months after (before) January 1, 2016. Dealer-specific control variables are lagged and include the change in the five-year CDS spread and the log volume of currency forwards. Fixed effects are denoted by *fe*. Standard errors (in parentheses) are clustered by dealer. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% level, respectively. The basis is constructed by synchronizing contract-level currency forwards between dealers and clients from DTCC Derivatives Repository with spot exchange rates and interest rates from Refinitiv Tick History. CDS spreads are from Bloomberg.

	(1)	(2)	(3)	(4)
<i>Affected Dealers</i> × <i>Post Regulation</i>	−18.680*** (5.580)	−18.112*** (5.893)	−15.896** (7.434)	−15.826** (7.801)
Δ <i>Dealer CDS</i>		−0.679 (0.394)		−0.467 (0.533)
<i>Dealer Log Volume</i>		−2.472 (3.085)		4.630** (2.094)
<i>R</i> ²	0.28	0.29	0.41	0.41
# <i>Observations</i>	25,538	25,538	22,293	22,293
<i>Dealer fe</i>	✓	✓	✓	✓
<i>Currency fe</i>	✓	✓		
<i>Client</i> × <i>Time fe</i>	✓	✓		
<i>Currency</i> × <i>Client</i> × <i>Time fe</i>			✓	✓

basis between dealers who were subject to the regulatory intervention (the treatment group) and dealers who were exempt (the control group) before and after the regulatory change. A negative estimate for γ would suggest that affected dealers (relative to unaffected dealers) charge clients borrowing dollars through the FX market a premium in response to the regulatory shock. We complement our specifications with both time-invariant (e.g., dealer) and time-variant (e.g., currency × client × time) fixed effects (*fe*), where the interaction with time is obtained with a weekly time (calendar date) fixed effect. We estimate the difference-in-differences regressions via least squares and cluster standard errors at the dealer level.¹¹

Table II reports the estimates of γ using different specifications with multiple combinations of fixed effects. We find that estimates of γ are always negative and statistically significant, even after controlling for observed dealer characteristics and absorbing changes in demand conditions at the currency

¹¹ Our results hold if we collapse data into a pre- and postregulatory period to ensure that the standard errors are not artificially low because of serial correlation (e.g., Bertrand, Duflo, and Mullainathan (2004)). We also run a wild cluster bootstrap exercise with 1,000 replications to verify the robustness of our analysis for the number of clusters (e.g., Cameron and Miller (2014)).

$\times$ client level. Our findings suggest that, after the introduction of the U.K. leverage ratio framework, dealer banks subject to the policy shock charged a wider negative dollar basis to their clients relative to unaffected dealer banks.

In specifications (1) and (2), for example, we control for time-invariant dealer and currency fixed effects as well as for time-variant unobserved characteristics at the client level in the spirit of Khwaja and Mian (2008). This is important as individual counterparty fundamentals could drive the widening of the dollar basis irrespective of dealer characteristics and we would be unable to give a supply interpretation to our estimates. Note that separate time fixed effects are spanned by client $\times$ time fixed effects and thus are omitted here. The first specification does not include dealer-specific variables and reports a negative and highly statistically significant estimate on γ of -18.68 bps per annum with a standard error of 5.58. The second specification adds controls for dealer-specific variables but yields qualitatively similar results: the estimate on γ is about -18.11 bps per annum with a standard error of 5.89. To translate these estimates into economic terms, we employ a simple back-of-the-envelope calculation. Since the average maturity of a forward contract is roughly 33 days while its average volume is about \$43.75 million in our sample, we find that, on average, a dealer bank affected by the regulatory shock will charge its client an additional cost of approximately $(18.11/10,000) \times (33/365) \times 43.75$ million = \$7,164 per transaction relative to unaffected banks.

Specifications (3) and (4) further absorb time-variant unobserved characteristics at the currency level via currency $\times$ client $\times$ time fixed effects. Such characteristics include, for example, heterogeneity in currency liquidity and volatility, which are not directly observable. The first of these specifications does not include dealer-specific control variables and produces a negative and statistically significant γ estimate of about -15.90 bps per annum with a standard error of 7.43. The second specification controls for dealer-specific variables and yields a virtually identical estimate. The economic value of γ remains large and translates into a premium of about \$6,260 per transaction between affected and unaffected dealer banks.

To summarize, we show that the supply of dollar funding in the FX market steepened for affected dealer banks while controlling for changes in client demand, after the introduction of the U.K. leverage ratio framework. We thus provide empirical evidence on the causal relationship between the leverage ratio requirement and deviations from the CIP condition.

C. Dollar Basis and Trading Direction

The negative dollar basis has been associated with an increase in FX hedging imbalances and tighter balance sheet costs for global banks (e.g., Du, Tepper, and Verdelhan (2018), Borio et al. (2018)). This observation begs the question of whether the leverage ratio requirement has a more pronounced effect on clients who synthetically borrow U.S. dollars (or equivalently sell U.S. dollars forward) as opposed to clients who synthetically lend U.S. dollars (or buy U.S.

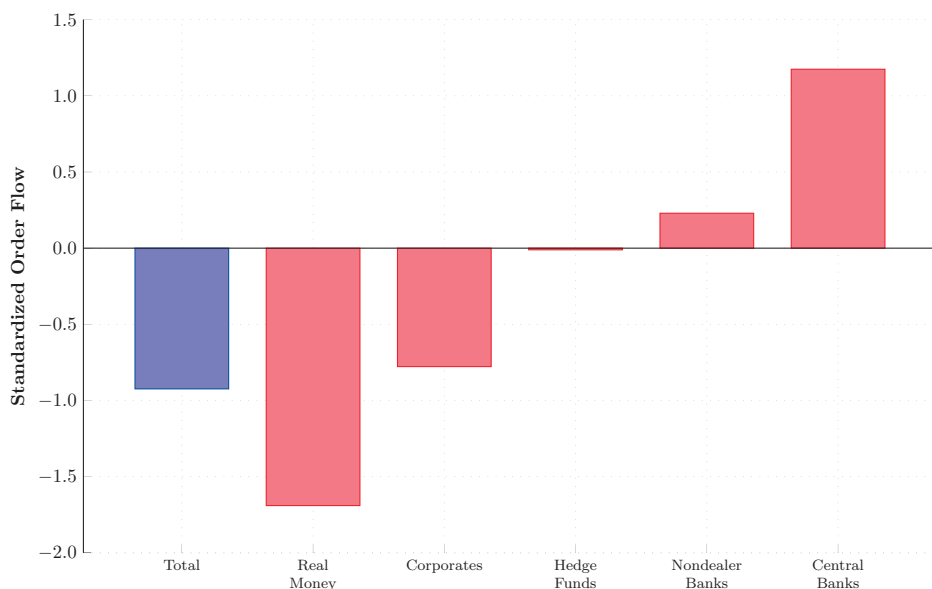


Figure 3. Order flow imbalances in the forward market. This figure displays the average order flow imbalance between (initiating) clients and dealers in the forward market. The order flow imbalance is the value of buyer-initiated orders minus the value of seller-initiated orders of U.S. dollars against foreign currencies such that a negative (positive) order flow indicates net selling (buying) pressure on the U.S. dollar in the forward market by end-user clients. The order flow is constructed using currency forwards between dealers and clients from the DTCC Derivatives Repository. For each sector (e.g., Real Money), we standardize the average order imbalance by its sample standard deviation for ease of comparison. The sample runs at the weekly frequency from November 2015 to December 2016. (Color figure can be viewed at wileyonlinelibrary.com)

dollars forward). As a preliminary exercise, we inspect the aggregate order imbalances between dealers and clients in the forward market, that is, the value of buyer-initiated orders minus the value of seller-initiated orders for a given currency pair with the initiator of the transaction being the nonquoting counterparty. We rely on the buy/sell indicator of currency forwards available in our data set between November 2015 and December 2016. Specifically, we sign a transaction negatively (positively) if the initiator of the transaction is selling (buying) dollars in the forward market against dealers such that a negative (positive) order flow indicates net selling (buying) pressure on the U.S. dollar in the forward market by end-user clients. Finally, we aggregate the order imbalance across all currency pairs and client sectors every week and then display the mean values, standardized by the corresponding standard deviation to ease the comparison, in Figure 3. A negative value means that, on average, clients in a given sector sell dollars forward or equivalently fund themselves in dollars in the FX market, and vice versa. When we categorize order flow by client type, we document that real money and nonfinancial corporate firms sell dollars forward, in line with recent literature. Arai et al. (2016), for example, report that

Japanese insurance companies hedge most of their foreign bond investments while corporations have also increased the issuance of bonds denominated in euro and yen while swapping these proceeds into dollars. Central banks and nondealer banks, in contrast, buy dollars forward, consistent with anecdotal evidence that some central banks have exploited CIP deviations for “return enhancement” on their FX reserves (e.g., Debelle (2017)) and nondealer banks likely acting on behalf of other investors. The average order flow for hedge funds is approximately zero, consistent with the fact that they typically bet on dollar-neutral long-short strategies such as carry, value, and momentum. Overall, our sample confirms the aggregate findings reported by recent literature.

We now address our question using the buy/sell indicator of currency forwards to categorize each client transaction by its trading direction. In doing so, we can unpack the left-hand side of equation (5) for client transactions selling and buying dollars forward against dealer banks. The former demand synthetic dollars whereas the latter supply synthetic dollars. Note that we do not change the sign of the dollar basis depending on the direction of the transaction but rather run separate difference-in-differences regressions for different types of client transactions.

We report the results in Table III and find that estimates of γ are always negative and statistically significant for clients selling dollars forward, even after controlling for the dealer-specific variables and absorbing changes in demand conditions at the currency $\times$ client level. In contrast, we record a slightly positive yet statistically insignificant estimate of γ for clients buying dollars forward. These findings indicate that, after the introduction of the U.K. leverage ratio framework, dealer banks subject to the policy shock charged a wider negative dollar basis only to clients seeking dollar funding in the FX market relative to unaffected dealer banks.¹²

Specification (1), for example, employs the dollar basis for clients selling dollars forward as the dependent variable. We add currency $\times$ client $\times$ time fixed effects that absorb all time-variant unobserved characteristics at the client $\times$ currency level beyond a time-invariant dealer fixed effect. The estimate of γ is -21.71 bps per annum and highly statistically significant with a standard error of 5.18. In terms of economic value, on average, dealer banks affected by the policy shock (relative to unaffected banks) charge clients selling dollars forward an additional $(21.71/10,000) \times (33.2/365) \times 48.69$ million = \$9,615 per transaction, given that the average maturity and volume of forwards are close to 33.2 days and 48.69 million, respectively, for this type of client in our sample. Specification (2) introduces dealer-specific control variables and the results remain both statistically and economically equivalent. Specification (3) uses the dollar basis for clients buying dollars forward as the left-side variable but produces a statistically insignificant estimate on γ of 2.70 bps per

¹² Table III employs fewer observations than Table II. While the latter table retains all clients with at least two trading relationships in a given week regardless of the trading direction, the former table only retains clients with at least two buying or two selling trading relationships in a given week.

Table III
Dollar Basis and Leverage Ratio: Trading Direction

This table presents difference-in-differences estimates associated with the introduction of the U.K. leverage ratio framework on January 1, 2016. The dependent variable is the volume-weighted dollar basis from currency forwards measured weekly at the currency, dealer, and client level in bps per annum. Selling (Buying) denotes client transactions that sell (buy) U.S. dollars forward. *Affected Dealers* is a dummy equal to one (zero) for major U.K. banks (subsidiaries of international banks). *Post Regulation* is a dummy equal to one (zero) for a posttreatment (pretreatment) period of two months after (before) January 1, 2016. Dealer-specific control variables are lagged and include the change in the five-year CDS spread and the log volume of currency forwards. Fixed effects are denoted by *fe*. Standard errors (in parentheses) are clustered by dealer. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% level, respectively. The basis is constructed by synchronizing contract-level currency forwards between dealers and clients from DTCC Derivatives Repository with spot exchange rates and interest rates from Refinitiv Tick History. CDS spreads are from Bloomberg.

	Selling		Buying	
	(1)	(2)	(3)	(4)
<i>Affected Dealers</i> × <i>Post Regulation</i>	−21.706*** (5.183)	−21.751*** (5.347)	2.704 (12.585)	2.434 (12.255)
Δ <i>Dealer CDS</i>		0.274 (0.531)		−0.034 (0.569)
<i>Dealer Log Volume</i>		−0.826 (3.848)		6.784 (7.172)
<i>R</i> ²	0.52	0.52	0.49	0.49
# <i>Observations</i>	8,842	8,842	8,875	8,875
<i>Dealer fe</i>	✓	✓	✓	✓
<i>Currency</i> × <i>Client</i> × <i>Time fe</i>	✓	✓	✓	✓

annum with a standard error of 12.59. In economic terms, this is equivalent to an additional cost of $(2.70/10,000) \times (32.9/365) \times 38.42 \text{ million} = \935 . Controlling for dealer-specific variables in specification (4) makes no qualitative difference.¹³ Figure 4 summarizes the impact of the regulatory intervention on the dollar basis for both clients selling (left chart) and clients buying (right chart) dollars forward. We show the average difference in dollar basis between treated and untreated dealer banks two months before and after the change in the leverage ratio reporting requirements. The shaded area denotes the 95% confidence interval. While for clients borrowing synthetic dollars we observe a statistically significant decline in the mean difference after the regulatory shock, we find no change in the mean difference for clients lending synthetic

¹³ Table IA.V in the Internet Appendix employs standard errors clustered by dealer and time (calendar week). Results remain largely comparable, albeit with slightly smaller standard errors. This may be due to the fact that double-clustered standard errors do not always yield unbiased estimates with a limited number of clusters (e.g., Petersen (2009), Thompson (2011)). In Table IA.VI, we employ a volume-weighted daily dollar basis but results remain largely comparable, notwithstanding a lower number of clients with multiple trading relationships on a single day.

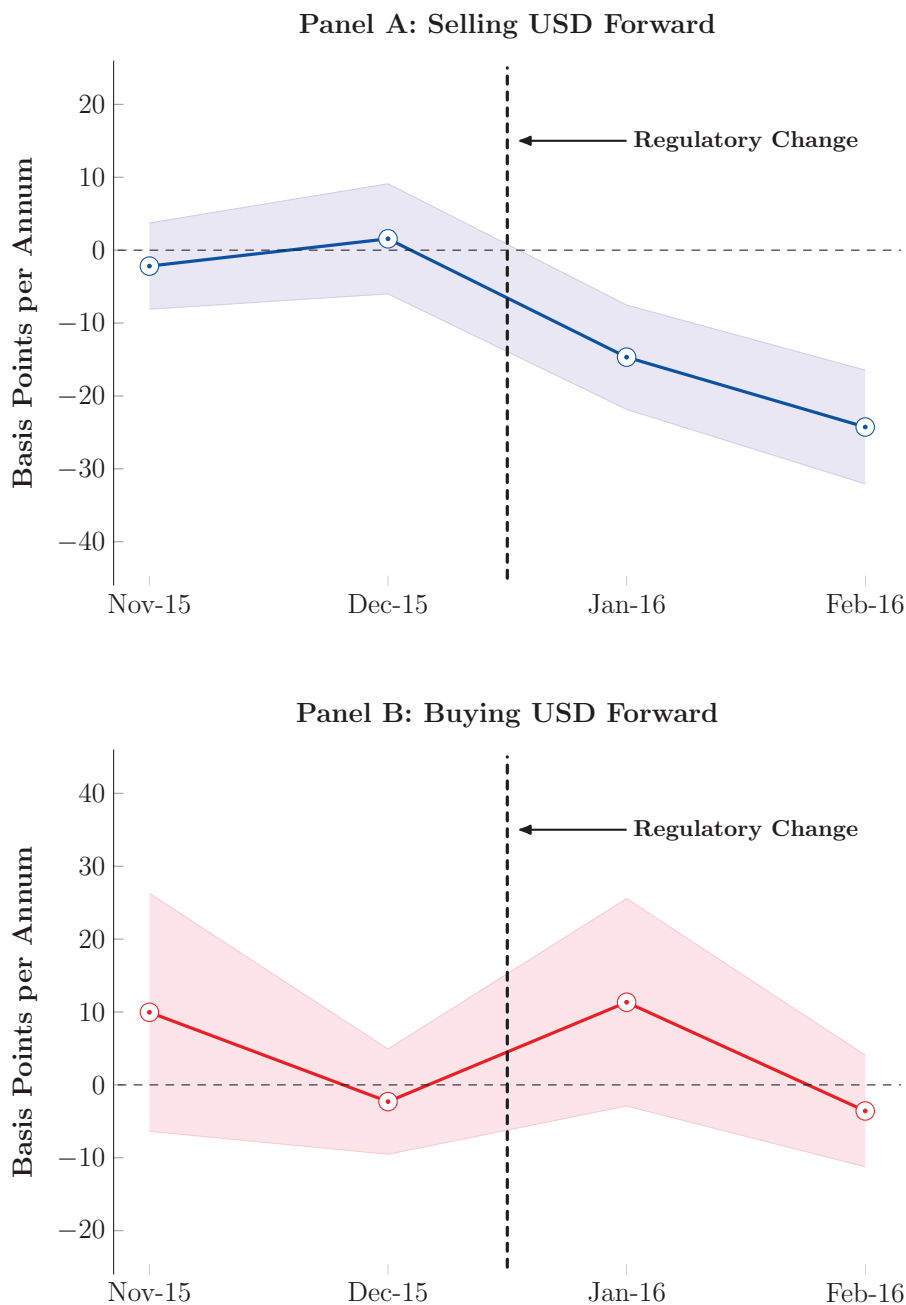


Figure 4. Dollar basis and leverage ratio. This figure displays the average difference in the volume-weighted dollar basis between *affected* and *unaffected* dealer banks two months before and after the introduction of the U.K. leverage ratio framework on January 1, 2016. The basis is constructed by synchronizing contract-level currency forwards between dealers and clients from DTCC Derivatives Repository with spot exchange rates and interest rates from Refinitiv Tick History. Selling (Buying) denotes client transactions that sell (buy) U.S. dollars forward. The basis is expressed in bps per annum and the shaded areas denote the 95% confidence intervals. (Color figure can be viewed at wileyonlinelibrary.com)

Table IV
Dollar Basis and Leverage Ratio: End-Users

This table presents difference-in-differences estimates associated with the introduction of the U.K. leverage ratio framework on January 1, 2016. The dependent variable is the volume-weighted dollar basis from currency forwards measured weekly at the currency, dealer, and client level in bps per annum. Hedgers (Others) refers to real money and nonfinancial corporates (hedge funds, nondealer banks, and central banks). *Affected Dealers* is a dummy equal to one (zero) for major U.K. banks (subsidiaries of international banks). *Post Regulation* is a dummy equal to one (zero) for a posttreatment (pretreatment) period of two months after (before) January 1, 2016. Dealer-specific control variables are lagged and include the change in the five-year CDS spread and the log volume of currency forwards. Fixed effects are denoted by *fe*. Standard errors (in parentheses) are clustered by dealer. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% level, respectively. The basis is constructed by synchronizing contract-level currency forwards between dealers and clients from DTCC Derivatives Repository with spot exchange rates and interest rates from Refinitiv Tick History. CDS spreads are from Bloomberg.

	Hedgers		Others	
	(1)	(2)	(3)	(4)
<i>Affected Dealers</i> × <i>Post Regulation</i>	−19.740*** (4.988)	−19.363*** (4.970)	−7.331 (5.336)	−8.188 (5.344)
Δ <i>Dealer CDS</i>		−0.977*** (0.256)		0.532** (0.241)
<i>Dealer Log Volume</i>		4.666* (2.606)		3.408 (2.400)
<i>R</i> ²	0.42	0.42	0.40	0.40
# <i>Observations</i>	14,539	14,539	7,754	7,754
<i>Dealer fe</i>	✓	✓	✓	✓
<i>Currency</i> × <i>Client</i> × <i>Time fe</i>	✓	✓	✓	✓

dollars. In Figure IA.6 in the Internet Appendix, we plot the average difference in dollar basis between treated and untreated dealer banks until May 2016 and find that the effect attenuates after about five months. A possible explanation could be that, other things being equal, clients of affected dealers migrated to unaffected dealers following the regulatory change. We investigate this hypothesis in Section IV.

In addition to using the buy/sell indicator, we group end-users into natural hedgers (i.e., real money investors and nonfinancial corporate firms) and others (i.e., hedge funds, nondealer banks, and central banks), and we run separate difference-in-differences regressions for both types of end-users. We display the results in Table IV. Estimates of γ for natural hedgers are largely comparable, both statistically and economically, with those for clients selling dollars forward in Table III. For the group of end-users labeled *others*, we obtain a negative but statistically insignificant estimate of γ . This test further confirms the effect of the leverage ratio requirement on the dollar basis for clients demanding dollar funding as opposed to clients providing dollar funding in the FX market.

D. Placebo Tests

The difference-in-differences method relies on the assumption that outcomes have equal trends between affected and unaffected dealers in the absence of treatment, that is, confounders varying across groups are time-invariant and confounders varying across time are group-invariant. When the underlying common trend assumption is invalid, the estimate of γ is biased as the trend for the control group is not a valid estimate of the counterfactual trend that we would have observed for the treatment group in the absence of the regulatory change. We test the validity of the underlying assumption of equal trends in two ways. In our first assessment, we run difference-in-differences regressions using a placebo regulatory date, that is, a fake shock at a different time than the actual regulatory change. In our second assessment, our empirical specification employs a placebo outcome, that is, a dollar basis implied from contracts that should not be affected by the introduction of the U.K. leverage framework. Table V presents estimates of γ for two placebo dates. Panel A employs April 1, 2016 as a placebo shock and uses the dollar basis for all client transactions, client transactions selling dollars forwards only, and only client transactions buying dollars forwards, respectively, as the dependent variable. Estimates of γ are not only statistically insignificant but also economically small. For example, for all client transactions, the estimate of γ corresponds to less than 3.4 bps per annum. Panel B considers October 1, 2015 as a placebo shock and uses the dollar basis for all client transactions, natural hedgers only, and other end-users only, as the dependent variable (recall that the buy/sell indicator is not available before November 2015 in our sample). Results remain both qualitatively and quantitatively similar.

Table VI displays estimates of γ for two placebo outcomes. Panel A uses the actual regulatory shock on January 1, 2016 but employs the dollar basis from currency forwards with intramonth contractual and maturity dates as the dependent variable. These forwards should not alter the month-end balance sheet capacity of dealer banks as they do not enter the leverage ratio calculation. We find that estimates of γ are statistically insignificant and economically small regardless of whether we use all client transactions or split them into transactions that buy versus sell dollars forward. For all client transactions, for example, the estimate of γ is about 3.2 bps per annum.

In Panel B, we continue to use the actual regulatory shock on January 1, 2016, but we employ the dollar basis implied from currency basis swaps as the outcome variable. Currency basis swaps are long-term contracts with a maturity of up to 30 years whose spread is quoted on the foreign currency leg against the dollar and can be interpreted as long-term CIP deviations (e.g., Du, Tepper, and Verdelhan (2018)).¹⁴ For this test, we use the spread of contract-level currency basis swaps measured at the currency $\times$ dealer $\times$

¹⁴ “Paying” the basis means borrowing the foreign currency versus lending dollars while “receiving” the basis implies lending the foreign currency versus borrowing in dollars. A negative basis denotes strong demand for dollars as one party is willing to receive a lower interest rate on its foreign currency position.

Table V
Dollar Basis and Leverage Ratio: Placebo Dates

This table presents difference-in-differences estimates associated with two placebo event dates. The dependent variable is the volume-weighted dollar basis from currency forwards measured weekly at the currency, dealer, and client level in bps per annum. Selling (Buying) denotes client transactions that sell (buy) U.S. dollars forward. Hedgers (Others) refers to real money and non-financial corporate firms (hedge funds, nondealer banks, and central banks). *Affected Dealers* is a dummy equal to one (zero) for major U.K. banks (subsidiaries of international banks). *Post Placebo Date* is a dummy equal to one (zero) for the posttreatment (pretreatment) period that covers two months after (before) the placebo date. Dealer-specific control variables are lagged and include the change in the five-year CDS spread and the log volume of currency forwards. We include dealer and currency $\times$ client $\times$ time (calendar date) fixed effects *fe*. Standard errors (in parentheses) are clustered by dealer dimension. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% level, respectively. The basis is constructed by synchronizing contract-level currency forwards between dealers and clients from DTCC Derivatives Repository with spot exchange rates and interest rates from Refinitiv Tick History. CDS spreads are from Bloomberg.

Panel A: Placebo Date on April 1, 2016			
	All	Selling	Buying
<i>Affected Dealers</i> $\times$ <i>Post Placebo Date</i>	3.398 (3.189)	5.116 (4.334)	3.378 (4.876)
<i>R</i> ²	0.50	0.50	0.49
# <i>Observations</i>	19,869	10,295	9,574
Panel B: Placebo Date on October 1, 2015			
	All	Hedgers	Others
<i>Affected Dealers</i> $\times$ <i>Post Placebo Date</i>	5.199 (4.806)	5.797 (6.261)	4.375 (6.747)
<i>R</i> ²	0.47	0.48	0.46
# <i>Observations</i>	17,377	11,527	5,850
<i>Control Variables</i>	✓	✓	✓
<i>Dealer fe</i>	✓	✓	✓
<i>Currency</i> $\times$ <i>Client</i> $\times$ <i>Time fe</i>	✓	✓	✓

sector level and saturate the difference-in-differences regressions with dealer and currency $\times$ sector $\times$ time fixed effects. Note that we cannot use weekly observations together with client fixed effects in this analysis as we observe few clients with multiple trading relationships at the weekly frequency. In addition, we present results for all transactions as well as for natural hedgers and other end-users, since the buy/sell indicator is unavailable for currency basis swaps. We find that the estimates of γ are statistically insignificant and economically small, which is not surprising as currency basis swaps affect the balance sheet capacity of dealer banks both at the month-end and the quarter-end. To summarize, we provide empirical evidence using both placebo shocks and placebo outcomes on the validity of the parallel trends assumption. These

Table VI
Dollar Basis and Leverage Ratio: Placebo Outcomes

This table presents difference-in-differences estimates associated with the introduction of the U.K. leverage ratio framework on January 1, 2016. The dependent variable, in bps per annum, is the volume-weighted dollar basis from currency forwards (with intramonth contractual and maturity dates) measured weekly at the currency, dealer, and client level in Panel A, and the contract-level dollar basis from currency basis swaps measured at the currency, dealer, and sector level in Panel B. Selling (Buying) denotes client transactions that sell (buy) the dollars forward. Hedgers (Others) refers to real money and nonfinancial corporate firms (hedge funds, nondealer banks, and central banks). *Affected Dealers* is a dummy equal to one (zero) for major U.K. (international) banks. *Post Regulation* is a dummy equal to one (zero) for a posttreatment (pretreatment) period of two months after (before) January 1, 2016. Dealer-specific control variables are lagged and include the change in the five-year CDS spread and the log volume of currency forwards or currency basis swaps. Fixed effects are denoted by *fe*. Standard errors (in parentheses) are clustered by dealer. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% level, respectively. In Panel A, the basis is constructed by synchronizing forwards between dealers and clients from DTCC Derivatives Repository with spot rates and interest rates from Refinitiv Tick History. In Panel B, the basis is directly available from DTCC Derivatives Repository. CDS spreads are from Bloomberg.

Panel A: Dollar Basis from Currency Forwards			
	All	Selling	Buying
<i>Affected Dealers</i> × <i>Post Regulation</i>	3.167 (6.531)	6.608 (8.097)	−2.213 (10.788)
<i>R</i> ²	0.64	0.59	0.67
# <i>Observations</i>	1,023	502	521
<i>Control Variables</i>	✓	✓	✓
<i>Dealer fe</i>	✓	✓	✓
<i>Currency</i> × <i>Client</i> × <i>Time fe</i>	✓	✓	✓
Panel B: Dollar Basis from Currency Basis Swaps			
	All	Hedgers	Others
<i>Affected Dealers</i> × <i>Post Regulation</i>	3.467 (3.139)	3.871 (6.668)	2.455 (3.216)
<i>R</i> ²	0.61	0.93	0.56
# <i>Observations</i>	1,742	164	1,578
<i>Control Variables</i>	✓	✓	✓
<i>Dealer/Maturity/Hour fe</i>	✓	✓	✓
<i>Currency</i> × <i>Sector</i> × <i>Time fe</i>	✓	✓	✓

analyses also mitigate concerns about the role of regulatory shocks that take place within our sample period.

E. Risk-Weighted Capital Requirements

In addition to the minimum leverage ratio that is independent of any risk assessment, dealer banks are subject to risk-weighted capital requirements,

which have increased substantially since the global financial crisis. The key feature of the capital requirements is that risk weights depend on both credit risk and market risk, with the latter measured using a VaR based on 10-day holding-period returns that are typically computed over a window that corresponds to the past calendar year. Since short-term arbitrage opportunities have zero VaR, the risk-weighted capital requirements are more likely to play a role for long-term arbitrage deviations (Du, Tepper, and Verdelhan (2018)). We complement our analysis above by investigating the impact of risk-weighted capital measures on both short-term and long-term arbitrage deviations. Since the introduction of the Basel I framework, regulators in most countries have set capital requirements at a fixed value of 8% of risk-weighted assets. In the United Kingdom, this is reflected in what the domestic regulator calls the Pillar 1 requirements, which are meant to capture credit risk and market risk. However, the U.K. regulator introduced an additional bank-specific buffer—the Pillar 2 requirements—to capture risks not accounted for in the Pillar 1 requirements, such as operational and legal risks. Crucially for our analysis, these additional Pillar 2 capital requirements differ across banks and over time at the discretion of the regulator.¹⁵ We therefore use the Pillar 2 capital requirements as the key explanatory variable in a panel regression setting to study the impact of bank-specific risk-weighted capital requirements on the dollar basis.

We run fixed-effects panel regressions between December 2014 and December 2016. In Panel A of Table VII, we focus on the (short-term) dollar basis implied from currency forwards using the specification

$$B_{ijk,t} = \beta C_{j,t-1} + \delta' X_{j,t-1} + fe + \varepsilon_{ijk,t}, \quad (6)$$

where $B_{ijk,t}$ is the volume-weighted dollar basis in week t for currency i , dealer j , and counterparty κ , $C_{j,t-1}$ is the Pillar 2 component of the risk-weighted capital requirements for dealer j , and $X_{j,t-1}$ is a set of bank-specific lagged control variables, namely, the change in the five-year CDS spread, log volume of currency forwards, bank size (log of a bank's total assets in levels), liquid asset share (holdings of liquid assets scaled by nonequity liabilities), and deposit share (fraction of the bank's balance sheet financed with core deposits).¹⁶ We saturate our specifications with both time-invariant (e.g., dealer) and time-variant (e.g., currency $\times$ client $\times$ time) fixed effects (fe). The interaction with time is obtained with calendar date (week) fixed effects. We cluster standard errors along the dealer and time dimensions. The estimates of β , across different specifications, are all statistically insignificant, suggesting

¹⁵ Aiyar, Calomiris, Hooley, Korniyenko, and Wieladek (2014) and Forbes, Reinhardt, and Wieladek (2017) argue that Pillar 2 requirements should reflect mostly nonbalance sheet risks and therefore should be regarded as exogenous to bank balance sheet variables.

¹⁶ The Pillar 2 of the risk-weighted capital requirements, bank size, liquid asset share, and deposit share are collected from the Bank of England. These data are available quarterly. We obtain higher-frequency observations by forward filling, that is, we retain the latest available observation until a new one is made available. We present a description of these variables in Panel B of Table IA.VII in the Internet Appendix.

Table VII
Dollar Basis and Capital Requirements

This table presents fixed-effects panel regression estimates. The dependent variable, in bps per annum, is the volume-weighted dollar basis from currency forwards measured weekly at the dealer, currency, and client level in Panel A, and the contract-level dollar basis from currency basis swaps measured at the dealer, currency, and sector level in Panel B. *Capital Requirements* denote bank-specific risk-weighted capital requirements set by the U.K. regulator. Dealer-specific control variables are lagged and include the change in the five-year CDS spread, log volume of forwards or currency basis swaps, bank size (log of total assets), liquid asset share (holdings of cash and market loans scaled by nonequity liabilities), and deposit share (fraction of the balance sheet financed with core deposits). Fixed effects are denoted by *fe*. Standard errors (in parentheses) are clustered by time and dealer. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% level, respectively. The sample runs from December 2014 to December 2016. In Panel A, the basis is constructed by synchronizing forwards between dealers and clients from DTCC Derivatives Repository with spot rates and interest rates from Refinitiv Tick History. In Panel B, the basis is directly available from DTCC Derivatives Repository. CDS spreads are from Bloomberg and other data from the Bank of England.

	(1)	(2)	(3)	(4)
Panel A: Dollar Basis from Currency Forwards				
<i>Capital Requirements</i>	0.314 (0.893)	0.298 (1.279)	1.132 (1.000)	0.994 (1.540)
R^2	0.39	0.39	0.49	0.49
# Observations	749,895	749,895	338,278	338,278
<i>Control Variables</i>		✓		✓
<i>Dealer fe</i>	✓	✓	✓	✓
<i>Currency fe</i>	✓	✓		
<i>Client × Time fe</i>	✓	✓		
<i>Currency × Client × Time fe</i>			✓	✓
Panel B: Dollar Basis from Currency Basis Swaps				
<i>Capital Requirements</i>	−4.482** (1.620)	−5.260*** (1.075)	−5.020*** (1.568)	−5.794*** (1.039)
R^2	0.66	0.68	0.71	0.73
# Observations	7,802	7,802	7,802	7,802
<i>Control Variables</i>		✓		✓
<i>Dealer / Maturity / Hour fe</i>	✓	✓	✓	✓
<i>Currency fe</i>	✓	✓		
<i>Sector × Time fe</i>	✓	✓		
<i>Currency × Sector × Time fe</i>			✓	✓

that the risk-weighted capital requirements have a negligible impact on the short-term dollar basis.

In Panel B, we turn to the (long-term) basis from currency basis swaps using specifications similar to those in equation (6). Specifically, we replace $B_{ijk,t}$ with $B_{ijk\ell,t}^{xccy}$, that is, the contract-level dollar basis from currency basis swaps recorded on day t for currency i , dealer j , counterparty κ , and maturity

ℓ . We saturate our specifications with both time-invariant (e.g., dealer, hour, maturity) and time-variant (e.g., currency $\times$ sector $\times$ time) fixed effects (fe). The interaction with time is obtained with calendar date (day) fixed effects. We cluster standard errors along the dealer and time dimensions. We find strong evidence that the risk-weighted capital requirements affect the long-term dollar basis, as the estimate of β is negative and highly statistically significant. For example, in specification (4), where we use the Pillar 2 capital requirements, we control for a variety of bank-specific variables, and we add both time-invariant (i.e., hour, maturity, and dealer) and time-variant (i.e., currency-sector-time) fixed effects, we find highly statistically significant estimate of -5.79 . In economic terms, the currency basis swap widens by approximately 18 bps per annum when the lagged capital requirements increase by one standard deviation (i.e., $5.79 \times 3.10 \approx 18$). Our results suggest that long-term CIP deviations, in contrast to short-term CIP deviations, have increased in response to higher risk-weighted capital requirements.¹⁷

F. Robustness Tests

We now present a battery of robustness tests that further corroborate our findings. To save space, we summarize the key findings here and report the detailed results in the [Internet Appendix](#).

F.1. Subsample Analysis by Currency and Maturity

In our main analysis, we focus on six currency pairs and currency forwards with maturity ranging between one week and three months. Table [IA.X](#) in the [Internet Appendix](#) investigates the robustness of our findings to different combinations of currency pairs and maturities. In Panel A, for example, we split the major currency pairs (i.e., EUR, GBP, and JPY relative to USD) from the other ones (i.e., AUD, CAD, and CHF relative to USD) and repeat the difference-in-differences regressions with trading directions. The estimate on γ for major currency pairs is highly statistically significant and virtually identical to the one documented in Table [III](#). For the remaining currency pairs, the impact of the leverage ratio on the dollar basis implied from γ is marginally more pronounced but comparable with the basket of major currency pairs. In Panel B, we separate short-maturity currency forwards (between one week and one month) from longer-maturity currency forwards. The magnitude of the γ estimate is slightly larger for short-term contracts (-25 with a standard error of 9.27) than long-term ones (-23 with a standard error of 6.92). These results suggest that the maturity of currency forwards (and the resulting annualization of the dollar basis) is not artificially driving the magnitude of

¹⁷ Results in Panel A remain qualitatively the same when using contract-level dollar basis and sector fixed effects. The estimates of δ for Panels A and B are presented in Tables [IA.VIII](#) and [IA.IX](#) in the [Internet Appendix](#).

our estimates. Overall, we confirm the general pattern uncovered in our main analysis. Table [IA.XI](#) in the [Internet Appendix](#) further constructs the dollar basis for standard maturity currency forwards (i.e., one week, one month, two months, and three months) and reports qualitatively similar results.

F.2. Client Hedging Demand

An alternative explanation could rationalize our findings. In particular, if for some unobserved reason U.K. banks absorbed a higher level of client hedging demand than international banks during the pretreatment period, then in the posttreatment period they may have counterbalanced a saturated balance sheet capacity by charging their clients a wider basis. To shed light on this concern, for the pretreatment period we regress the weekly order imbalance of dealer j and currency i on a dummy variable equal to one for affected dealer banks and zero otherwise. We also add the lagged dealer-specific control variables and currency $\times$ time fixed effects. We report the estimates in Table [IA.XII](#) in the [Internet Appendix](#) and find no statistically significant difference between affected and unaffected banks. As a result, heterogeneous hedging demand pressure from clients does not explain our main findings.

F.3. Effective Spread on Currency Forwards

The sections above provide evidence that the leverage ratio requirement affects the dollar basis when clients demand synthetic dollar funding. To corroborate this finding, we examine the impact of the leverage ratio requirement on the relative effective spread between dealers and clients. If dealers are less aggressive in seeking client orders, say for volatility, liquidity, or other unobserved reasons, we should expect dealers to charge a spread to clients regardless of their trading direction.

To test this conjecture, we repeat the difference-in-differences regressions presented in Table [III](#) but replace the dollar basis with the relative effective spread of contractual forward rates relative to interdealer quotes described in Section [II.C](#). We report the estimates in Table [IA.XIII](#) in the [Internet Appendix](#) and find evidence that the leverage ratio rule affects the effective spread when clients raise dollar funding via the FX market. The estimate of γ is approximately 2.84 and highly statistically significant, after controlling for dealer and currency-client-time fixed effects. Moreover, adding dealer-specific control variables does not qualitatively alter our estimate. In contrast, we find no statistically significant relationship between the effective spread and the leverage ratio requirement for clients trading on the opposite side of the FX market. Table [IA.XIV](#) in the [Internet Appendix](#) reports comparable results when using different currency and maturity combinations. Taken together, these results suggest that our main findings are unlikely to be attributed to liquidity effects.

F.4. The Public Disclosure of the Leverage Ratio

Next, as an alternative plausibly exogenous shock, we exploit the obligation of dealer banks to publicly disclose their point-in-time quarter-end leverage ratio after January 2015 (see, for example, Du, Tepper, and Verdelhan (2018), Haynes, McPhail, and Zhu (2019)). While the leverage ratio rule was not mandatory at the time, banks had an incentive to adjust their leverage ratio around the expected minimum requirement suggested by regulators. Failing to disclose a leverage ratio in line with these expectations could have increased, for example, the cost of capital for highly leveraged banks as a higher level of financial disclosure reduces information asymmetry (Haynes, McPhail, and Zhu (2019)).

Depending on their ex ante leverage exposure, banks with a larger shortfall in regulatory capital may have charged their counterparty clients a wider dollar basis. We construct the leverage ratio of dealer banks as of December 2007, that is, before the policy debate on excessive leverage in the banking sector (e.g., Draghi (2008), using shareholders' claims to total assets as provided by the Bank of England (see, for more details, Bank of England (2015a)). We then form a treatment group and a comparison group of dealer banks using the cross-sectional median value of this measure, with the former (latter) group comprising banks with a leverage ratio below (above) the median value. We then run difference-in-differences regressions around January 2015 to test the causal impact of the leverage ratio disclosure on the dollar basis. The public disclosure requirement constitutes a shock under the assumption that it was not anticipated by banks.¹⁸

In our test, the preintervention period runs from December 1, 2014 (when our sample starts) to December 19, 2014, while the postintervention period runs from January 12, 2015 to March 29, 2015. Both periods are intended to capture a month (quarter) of observations before (after) the regulatory shock while removing two weeks of data bracketing the end of the year. We discard forward contracts with intraquarter contractual and maturity dates as these contracts do not affect the leverage ratio, calculated as a point-in-time estimate at quarter-ends. In addition to using contracts relative to all clients, we categorize end-users by trading directions. Since the buy/sell indicator for forward contracts is not available for this period, we split clients into natural hedgers (i.e., real money investors and nonfinancial corporate firms) and others (i.e., hedge funds, nondealer banks, and central banks). The former (latter) are likely to demand (provide) synthetic dollar funding via the FX market.

We run difference-in-differences regressions and report the estimates of γ in Panel A of Table IA.XV in the Internet Appendix. For the natural hedgers, the estimate of γ is negative and highly statistically significant at -25 bps per

¹⁸ Results remain qualitatively the same if we group dealer banks using the leverage ratio as of December 2012, that is, before the proposal made by the Basel Committee on the introduction of a leverage ratio requirement (see Table IA.XVI in the Internet Appendix).

annum with a standard error of 9.56. For other end-users, we uncover no statistically significant estimate of γ . In Panel B, we use a placebo shock by shifting the time window by three months. We uncover no statistically significant estimate of γ . This alternative quasi-experimental exercise confirms the role of the leverage ratio requirement as a key driver of the negative dollar basis.¹⁹

IV. Further Analysis and Extensions

In this section, we present additional analyses that complement and refine the results reported earlier. First, we explore the role of bilateral netting between dealers and clients and investigate whether dealers affected by the U.K. leverage ratio framework have discriminated their clients on these terms. Second, we study whether clients with high exposure to affected dealers were able to switch to unaffected dealers after the regulatory change.

A. Netting between Dealers and Clients

A derivative contract generates an exposure originating from the underlying asset and an exposure reflecting the creditworthiness of the counterparty entity. The Basel III leverage ratio framework recognizes both types of exposures and proscribes a dealer from netting out long and short positions across underlying exposures unless eligible transactions are undertaken against the same counterparty. When netting is permitted, a bank can consolidate outstanding derivative positions against a particular counterparty into a single legal obligation. This means that highly offsetting transactions should have a quantitatively smaller impact on the balance sheet exposure of a dealer through the calculation of the leverage ratio. We are unable to precisely evaluate the amount of netted positions between dealers and clients as part of the relevant information is not directly available to us.²⁰ However, we identify clients who do not net transactions and clients who net transactions during the pretreatment period. We then test whether U.K. banks have charged them a different dollar basis after January 2016. For simplicity, we refer to clients with zero netting as directional clients and to clients with nonzero netting as nondirectional clients. In our sample, roughly 69% of the clients have zero netting against dealer banks during the pretreatment period.

¹⁹ In a related exercise reported in Table IA.XVII in the [Internet Appendix](#), we show that U.K. dealer banks with a low leverage ratio as of December 2014, that is, prior to the public disclosure requirement and the consultation process for the implementation of the U.K. leverage ratio framework initiated by the Prudential Regulation Authority, charge clients seeking synthetic dollar funding a spread larger than 40 bps per annum after January 2016 relative to other dealer banks.

²⁰ To assess the exposure arising from a derivative, a bank must consider the replacement cost computed via marking to market plus an add-on for potential future exposure. The treatment of these components, moreover, can be affected by the existence of an eligible bilateral or multilateral netting contract as well as any posted cash collateral. We provide more details on netting in [Internet Appendix Section III](#).

We incorporate the role of netting into our analysis by augmenting the specification presented in equation (5) as follows:

$$B_{ijk,t} = \gamma_1 D_{affected} \times D_{post} + \gamma_2 D_{affected} \times D_{post} \times D_{zeronet} + \gamma_3 D_{affected} \times D_{zeronet} + \gamma_4 D_{post} \times D_{zeronet} + \delta' X_{j,t-1} + \alpha_j + \alpha_{i,\kappa,t} + \varepsilon_{ijk,t}, \quad (7)$$

where $D_{zeronet}$ is a dummy that is equal to one when netting between dealer j and client κ is zero during the pretreatment period (and zero otherwise), α_j denotes the time-invariant dealer fixed effects, and $\alpha_{i,\kappa,t}$ denotes the currency $\times$ client $\times$ time fixed effects. The coefficient γ_1 captures the dollar basis between affected dealers and nondirectional clients whereas the sum of γ_1 and γ_2 quantifies the dollar basis between affected dealers and directional clients. Panel A of Table VIII reports the estimates of γ_1 and γ_2 . We document a substantial amount of heterogeneity between directional and nondirectional clients for synthetic dollar borrowing, as the estimate of γ_1 is about -6 bps (with a standard error of 5.7) whereas the estimate of γ_2 is close to -48 bps (with a standard error of 15). Taken together, these figures suggest that dealer banks affected by the regulatory shock have charged a premium of 54 (6) bps per annum to more (less) capital-intensive client transactions relative to unaffected dealer banks after January 2016. In other words, borrowing synthetic dollars is significantly more expensive for directional clients than for nondirectional clients as the former undertake transactions that erode the regulatory capital of a dealer bank. For synthetic dollar lending, in contrast, we register no statistical evidence of CIP deviations even after considering client netting.

In Panel B of Table VIII, we extend the analysis to the log volume of currency forwards by replacing $B_{ijk,t}$ with $\ln V_{ijk,t}$ in equation (7). For synthetic dollar funding, we find that major U.K. banks (relative to international banks) have traded substantially more with nondirectional clients and considerably less with directional clients. In particular, the estimates of γ_1 and γ_2 , both statistically significant, are about 0.25 and -0.48 , respectively, which translate into a 25% volume increase for less capital-intensive transactions and a 23% volume contraction for more capital-intensive transactions. For synthetic dollar lending, no statistical relationship between volume and regulatory change is recorded. We also offset long and short forward transactions between dealers and clients, regardless of the currency pair, but find qualitatively identical results as shown in Internet Appendix Table IA.XVIII. To summarize, the dollar basis widens and volume falls for more capital-intensive client transactions that synthetically borrow dollars against major U.K. banks (relative to other banks) during the posttreatment window.²¹

²¹ For log volume, we also run specifications without $D_{zeronet}$ and report the estimates in Table IA.XIX in the Internet Appendix. While estimates of γ are statistically insignificant, the sign suggests that major U.K. banks (relative to other banks) have traded less (more) with clients borrowing (lending) synthetic dollars. These results, however, confirm that incorporating client netting is critically important.

Table VIII
Client Netting and Leverage Ratio

This table presents difference-in-differences estimates associated with the introduction of the U.K. leverage ratio framework on January 1, 2016. The dependent variable is the volume-weighted dollar basis (log volume) of currency forwards in Panel A (Panel B) measured weekly at the currency, dealer, and client level in bps per annum (billions of U.S. dollars). Selling (Buying) denotes client transactions that sell (buy) U.S. dollars forward. *Affected Dealers* is a dummy equal to one (zero) for major U.K. banks (subsidiaries of international banks). *Post Regulation* is a dummy equal to one (zero) for a posttreatment (pretreatment) period of two months after (before) January 1, 2016. *Zero Netting* is a dummy equal to one (zero) when dealer-client netting (currency specific) during the pretreatment period is zero (nonzero). Dealer-specific control variables are lagged and include the change in the five-year CDS spread and the log volume of currency forwards. Fixed effects are denoted by *fe*. Standard errors (in parentheses) are clustered by dealer. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% level, respectively. The basis is constructed by synchronizing forwards between dealers and clients from DTCC Derivatives Repository with spot rates and interest rates from Refinitiv Tick History. In Panel B, the basis is directly available from DTCC Derivatives Repository. CDS spreads are from Bloomberg.

	All	Selling	Buying
Panel A: Dollar Basis			
<i>Affected Dealers</i> × <i>Post Regulation</i> × <i>Zero Netting</i>	−30.052*** (8.704)	−47.803*** (14.999)	7.166 (10.552)
<i>Affected Dealers</i> × <i>Post Regulation</i>	−5.717 (3.737)	−5.970 (5.684)	−2.639 (4.554)
<i>R</i> ²	0.44	0.55	0.52
# <i>Observations</i>	22,293	8,842	8,875
Panel B: Log Volume			
<i>Affected Dealers</i> × <i>Post Regulation</i> × <i>Zero Netting</i>	−0.143 (0.130)	−0.482*** (0.164)	0.192 (0.182)
<i>Affected Dealers</i> × <i>Post Regulation</i>	0.126* (0.074)	0.248** (0.097)	0.002 (0.101)
<i>R</i> ²	0.74	0.80	0.77
# <i>Observations</i>	22,293	8,842	8,875
<i>Control Variables and Other Interactions</i>	✓	✓	✓
<i>Dealer fe</i>	✓	✓	✓
<i>Currency</i> × <i>Client</i> × <i>Time fe</i>	✓	✓	✓

B. Substitution Effects and New Relationships

The findings reported in the previous section beg the question of whether directional clients (with more capital-intensive transactions and higher costs of synthetic dollar borrowing) were able to migrate from affected to unaffected dealer banks after the introduction of the leverage ratio framework. We address this question by running a set of client-level regressions that test whether directional clients who were more exposed to dealer banks affected by the new regulation have experienced a decline in their trading volume of

currency forwards relative to directional clients less exposed. In this exercise, we measure the average exposure of each client κ to affected dealers before the regulatory change using the ratio of client κ 's volume with affected dealers relative to client κ 's total volume during the preintervention period. In particular, we construct a dummy variable $D_{highexp}$ for each client κ that is equal to one if the same client has an above-median share of currency forwards intermediated by affected dealers, and zero otherwise. We then run client-level regressions based on the specification

$$\begin{aligned} \ln V_{i,j\kappa,t} = & \gamma_1 D_{highexp} \times D_{post} + \gamma_2 D_{highexp} \times D_{post} \times D_{zeronet} \\ & + \gamma_3 D_{highexp} \times D_{zeronet} + \gamma_4 D_{post} \times D_{zeronet} + \gamma_5 D_{highexp} + \gamma_6 D_{post} \\ & + \gamma_7 D_{zeronet} + \delta' X_{j,t-1} + \alpha_j + \hat{\alpha}_{i,\kappa,t} + \varepsilon_{i,j\kappa,t}, \end{aligned} \quad (8)$$

where γ_1 captures the impact of the regulatory change on the volume of nondirectional clients with high exposure, and the sum of γ_1 and γ_2 quantifies the impact of the regulatory change on the volume of directional clients with high exposure. Since we examine trading volume at the client level, we are unable to directly absorb changes in the client-currency demand conditions via time-varying fixed effects. We instead add the preestimate $\hat{\alpha}_{i,\kappa,t}$ from equation (7) to our specification in the spirit of Abowd, Kramarz, and Margolis (1999), Cingano, Manaresi, and Sette (2016), and Kotidis and van Horen (2018). Panel A of Table IX reports the estimates of γ_1 and γ_2 , which are statistically significant but have opposite signs for synthetic dollar borrowing, that is, $\gamma_1 = 0.122$ and $\gamma_2 = -0.240$. These estimates suggest that, in aggregate, directional clients with high exposure have experienced an 11.8% decline in their trading volume of currency forwards after the regulatory change. Recall that the estimates of γ_1 and γ_2 reported in Panel B of Table VIII are substantially larger in absolute terms and imply a 23% decrease in the trading volume of currency forwards between directional clients and affected dealers. Taken together, these figures suggest that international banks have received more transactions from directional clients with high exposure to major U.K. banks after the regulatory shock in January 2016, meaning that directional clients have migrated to some extent from affected to unaffected dealer banks. We also find that the trading volume of currency forwards has increased by roughly 12% (given the estimate of γ_1) for nondirectional clients with high exposure. Compared with the corresponding estimate in Panel B of Table VIII, we conclude that major U.K. banks have received more transactions from nondirectional clients after January 2016. Finally, for client transactions buying dollars forward, we record no statistical evidence of any substitution effect between major U.K. banks and international banks.

We further check whether substitution between affected and unaffected dealers took place by strengthening preexisting relationships or starting new relationships. Similar to Cingano, Manaresi, and Sette (2016) and Kotidis and van Horen (2018), we construct a dummy variable $NewRelationship_{i\kappa}$ that is equal to one (and zero otherwise) if client κ has a relationship with a new

Table IX
Substitution Effects and New Relationships

This table presents difference-in-differences estimates associated with the introduction of the U.K. leverage ratio framework on January 1, 2016. In Panel A, the dependent variable is the log volume of forwards measured weekly at the currency, dealer, and client level in billions of U.S. dollars. In Panel B, the dependent variable is a dummy variable equal to one (and zero otherwise) if a given client has a relationship with a new unaffected dealer after January 2016. *Selling* (*Buying*) denotes client transactions that sell (buy) U.S. dollars forward. *Post Regulation* is a dummy equal to one (zero) for a posttreatment (pretreatment) period of two months after (before) January 1, 2016. *High exposure* is a dummy equal to one (zero) if a client's volume to affected dealers relative to her total volume during the pretreatment period is above (below) the median value. *Zero Netting* is a dummy equal to one (zero) when dealer-client netting during the pretreatment period is zero (positive). Dealer-specific control variables are lagged and include the change in the five-year CDS spread and the log volume of currency forwards. Fixed effects are denoted by *fe*. Standard errors (in parentheses) are clustered by client. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% level, respectively. Volume is from currency forwards between dealers and clients from DTCC Derivatives Repository. CDS data are from Bloomberg.

	All	Selling	Buying
Panel A: Substitution Effects			
<i>Post Regulation</i> × <i>High Exposure</i> × <i>Zero Netting</i>	−0.080 (0.072)	−0.240** (0.095)	0.030 (0.108)
<i>Post Regulation</i> × <i>High Exposure</i>	0.051*** (0.018)	0.122*** (0.027)	−0.002 (0.030)
<i>R</i> ²	0.74	0.79	0.77
# <i>Observations</i>	22,293	8,842	8,875
<i>Dealer fe, Controls, and Other Regressors</i>	✓	✓	✓
<i>Currency</i> × <i>Client</i> × <i>Time</i> (Pre-estimated)	✓	✓	✓
Panel B: New Relationships			
<i>High Exposure</i> × <i>Zero Netting</i>	−0.004 (0.016)	−0.012 (0.020)	−0.016 (0.021)
<i>High Exposure</i>	0.006 (0.022)	0.019 (0.026)	0.027 (0.030)
<i>R</i> ²	0.69	0.72	0.65
# <i>Observations</i>	1,845	1,363	1,392
<i>Other Regressors</i>	✓	✓	✓
<i>Currency</i> × <i>Client</i> (Pre-estimated)	✓	✓	✓

unaffected dealer after January 2016, that is, trades during the postintervention period with an unaffected dealer j that was not among her counterparties during the preintervention period. Specifically, we run a client-level linear probability model following

$$NewRelationship_{ik} = \gamma_1 D_{highexp} + \gamma_2 D_{highexp} \times D_{zeronet} + \gamma_3 D_{zeronet} + \hat{\alpha}_{i,k} + \varepsilon_{jk}, \quad (9)$$

where γ_1 measures the probability that high-exposure nondirectional clients seek trading substitution with a new unaffected dealer, the sum of γ_1 and γ_2 indicates whether this occurs for high-exposure directional clients, and $\hat{\alpha}_{i,k}$ denotes preestimate currency $\times$ client fixed effects obtained by reestimating the specification in equation (7). We report the estimates of γ_1 and γ_2 in Panel B of Table IX and find no statistically significant evidence that more exposed clients (both directional or nondirectional) were more likely to establish a new relationship after January 2016. Our findings therefore suggest that clients with more capital-intensive transactions and high exposure to major U.K. banks during the preintervention period have increased trading activity in the postintervention period with unaffected dealers with which they already had a preexisting relationship rather than switch to new dealers.²²

V. Conclusions

The FX market—the largest financial market by daily turnover—has recently experienced large and persistent arbitrage deviations from the CIP condition, a simple no-arbitrage condition that simultaneously links together the spot/forward exchange rate markets with domestic/foreign money markets. While there is some evidence that attributes the existence of a negative dollar basis to dealers' balance sheet costs and hedging demand pressure (e.g., Borio et al. (2018), Du, Tepper, and Verdelhan (2018)), the relationship between the leverage ratio requirement and the dollar basis at the dealer level has not been tested directly. This paper fills this gap in the literature by directly examining the causal relationship between the leverage ratio requirement and the dollar basis. We employ a confidential data set on contract-level FX derivatives whose richness in terms of counterparty information is attractive from an identification perspective. We find that the additional funding cost international investors face in borrowing synthetic dollars through the FX market can be attributed to the leverage ratio requirement of dealer banks, and thus can be thought of as an unintended consequence of the postcrisis financial regulation. We also document a decrease in the volume of synthetic dollar borrowing, especially for clients who have zero netting against counterparty dealers. These are more capital-intensive transactions and thus would require that a dealer pledge an additional amount of regulatory capital. Overall, our findings help shed light on the potential costs of the leverage ratio framework. We leave any assessment of the net welfare effect of the leverage ratio rule for future research. Further research could also try to shed light on the question of whether market participants will adapt to the new regulation.

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²² In Internet Appendix Section IV, we also examine whether affected dealers stopped trading with existing clients and reduced their intake of new clients after the regulatory shock.

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Supporting Information

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Appendix S1: Internet Appendix.
Replication Code.